Stillness, Predictive Stability, and Causal Simulation

An Operational Formulation Independent of Numerical Units

25 October 2026 — revised formulation

Abstract

We replace uncalibrated latent depth scores with controlled observable laws. Histories are identified when no admissible future experiment can distinguish them. We prove that this predictive equivalence supports well-defined state updates, that exact seed cost is invariant under bijective recoding, and that predictive stability need not imply a small randomness requirement. Stillness is proposed as an empirically testable combination of low reported thought content, stable predictions under neutral continuation, and preserved stimulus discrimination. The causal Shannon bound applies conditionally to each specified finite experiment. No new empirical discovery or priority claim is asserted.

1 The experimental law comes first

Fix finite nonempty action and outcome alphabets $\mathcal{A}, \mathcal{Y}$. Let h be any finite action–outcome history and specify

$$p(y \mid h, a) > 0, \qquad \sum_{y} p(y \mid h, a) = 1.$$

These laws are a model to be estimated and tested, not consequences of a subjective description. Outcomes may jointly encode EEG summaries, behavior and reports, including missing responses. Actions include probes and a specified neutral continuation a_0. Positivity is an explicit restriction used below; zero branches require a separate formulation. No stationary or finite hidden-state assumption is imposed.

For each finite horizon k, a deterministic continuation policy maps the subsequent relative history to the next action. Let P_h^π denote its future outcome law starting after h. Its probabilities are products of the specified conditional kernels. The class of continuation policies and outcome coordinates is the same for every history compared.

2 Predictive states rather than arbitrary scores

Definition 1 (Predictive equivalence and distance)**.** *For histories h, h' define*

$$h \sim h' \quad \Longleftrightarrow \quad P_h^\pi = P_{h'}^\pi \quad \textit{for every finite horizon and continuation policy.}$$

For a fixed horizon $k \geq 1$, define

$$d_k(h, h') = \max_{\pi} \|P_h^\pi - P_{h'}^\pi\|_{\mathrm{TV}}, \qquad \|P - Q\|_{\mathrm{TV}} = \frac{1}{2} \sum_{z} |P(z) - Q(z)|.$$

Each d_k is a pseudometric: symmetry and the triangle inequality follow for each policy and survive taking the maximum. Moreover $h \sim h'$ exactly when $d_k(h, h') = 0$ for every k. Finite experimental data can test finite-horizon approximations, not establish this universal equality without additional assumptions.

Proposition 1 (Well-defined predictive dynamics)**.** *If $h \sim h'$, then*

$$p(y \mid h, a) = p(y \mid h', a), \qquad hay \sim h'ay$$

for every a, y. Hence the update $([h], a, y) \mapsto [hay]$ and the next-outcome kernel are well defined on predictive equivalence classes.

Proof. One-step experiments give equality of the next-outcome probabilities. Fix a, y and any continuation policy ρ. Construct a policy that first chooses a and, after y, follows ρ; elsewhere choose actions arbitrarily. Equality of its joint outcome laws gives equality on events with first outcome y. Divide by the common positive probability of y. The conditional future laws agree for every ρ and horizon, proving the successor equivalence. □

This quotient removes distinctions with no predictive consequence for the chosen interface. It need not be finite, and is not asserted to be a unique physical substrate or a model of consciousness itself.

3 Units and exact causal seed cost

Truncate the experimental law after n actions to obtain a finite tree T_n. A simulator has a finite seed U with one distribution for all policies and a deterministic decoder

$$Y_t = D(U, H_t, A_t).$$

Shared histories and actions must receive identical responses from a given seed atom. Let $C(T_n)$ be the infimum of $\mathrm{H}(U)$ among simulators exact under all deterministic adaptive policies.

Proposition 2 (Recoding invariance)**.** *Let T_n' be obtained by a probability-preserving rooted-tree isomorphism that bijectively recodes actions and outcomes, allowing the recodings to depend on the preceding history. Then*

$$C(T_n') = C(T_n).$$

Predictive distances are also preserved when compared experiments are transported through a common bijective recoding of their interfaces.

Proof. Translate a recoded history and action back to the original tree, apply its decoder, and translate its output forward. This uses the same seed and preserves shared-history consistency and exactness. The inverse isomorphism gives the reverse inequality for the infima. For distances, the common recoding bijects policies and outcome paths; total variation is unchanged by a bijection. □

Changing a numerical unit is such a recoding only when the conversion is exactly invertible on the recorded values. Rounding, coarser bins, and a different sampling schedule generally change the experiment. Invariance under recoding holds for agitated processes too: it is not a diagnostic of stillness.

Conditional Shannon bridge. For each policy set $I_\pi = -\log_2 P_\pi(Y_{0:n-1})$, $F_\pi(z) = \Pr(I_\pi \le z)$ and

$$L(T_n) = \int_0^\infty [1 - \min_\pi F_\pi(z)]\, dz.$$

Taking the positive finite-tree causal Shannon theorem of the companion development [1] as a premise gives a finite greedy seed U_G:

$$\boxed{L(T_n) \le C(T_n) \le \mathrm{H}(U_G) \le L(T_n) + c_\star, \qquad c_\star = \frac{1 + \log_2 e}{2}.}$$

Application is immediate because T_n has finite nonempty alphabets, finite depth and positive kernels. The theorem itself is not independently audited here. Recoding also preserves $L(T_n)$: the policy correspondence preserves every transcript atom probability.

4 A testable candidate for stillness

Fix an observational event $Q \subseteq \mathcal{Y}$ representing a report of low thought content. Fix a response readout $b : \mathcal{Y} \to \mathcal{B}$ and two specified probe actions a_+, a_-. Write $p_h^a = p(\cdot \mid h, a)$ and define

$$q(h) = p_h^{a_0}(Q), \qquad r(h) = \|b_\# p_h^{a_+} - b_\# p_h^{a_-}\|_{\mathrm{TV}},$$

$$s_k(h) = \max_{y \in \mathcal{Y}} d_k(h, ha_0y).$$

Here $b_\#$ denotes the induced law of the behavioral readout. The quantity r measures discrimination between probes, not consciousness or wakefulness by itself.

Definition 2 (Operational candidate)**.** *Given pre-specified tolerances η, ϵ and criterion $r_0 > 0$, call a history a candidate for stable responsive quiet when*

$$q(h) \geq 1 - \eta, \qquad s_k(h) \leq \epsilon, \qquad r(h) \geq r_0.$$

If wakefulness is part of the empirical claim, it must additionally be assessed using an independently specified measurement protocol.

All three quantities are dimensionless and preserved under consistent bijective recoding of the actions, outcomes, event Q, and readout b. Their experimental meaning still depends on the chosen probes, readout, time horizon, and tolerances. Dimensionless does not mean universal. In particular, neutral continuation is an experimental condition, not an assumption that the brain stops evolving.

Proposition 3 (Finite accumulation of predictive drift)**.** *If $h_j = h_{j-1}a_0y_j$ for $j = 1, \ldots, m$ and $s_k(h_{j-1}) \leq \epsilon_j$, then*

$$d_k(h_0, h_m) \leq \min\{1, \epsilon_1 + \cdots + \epsilon_m\}.$$

Proof. Each successive distance is at most ϵ_j by definition. Use the triangle inequality and the fact that total variation is at most one. □

This is a finite-horizon robustness bound, not a claim of permanent stability. The assumptions must hold along the continuation in question.

5 Stable does not mean free: an exact counterexample

Let actions be $0, 1$ and each outcome consist of two binary variables (R, B). Independently at each step conditional on the chosen action, let

$$\Pr(R = 0) = 1 - \eta, \qquad \Pr(B = 1 \mid a = 0) = p, \qquad \Pr(B = 1 \mid a = 1) = 1 - p,$$

where $0 < \eta < 1/2$ and $0 < p < 1/2$, and R, B are conditionally independent. Interpret $R = 0$ as a quiet report, take $a_0 = 0$, and use B as the response readout. All four outcomes have positive probability.

The kernel has no history dependence. Consequently all histories are predictively equivalent, $s_k(h) = 0$ for all k, $q(h) = 1 - \eta$, and $r(h) = 1 - 2p$. Nevertheless the exact seed cost is

$$\boxed{C(T_n) = n\,[h_2(\eta) + h_2(p)] > 0,}$$

where h_2 is binary entropy.

Proof. The constant-zero policy produces independent pairs with entropy $n[h_2(\eta) + h_2(p)]$. Its transcript is a deterministic function of any exact seed, giving the lower bound. For the upper bound, take independent pairs (R_t, Z_t) with $R_t \sim \text{Bernoulli}(\eta)$ and $Z_t \sim \text{Bernoulli}(p)$ as one seed. Output (R_t, B_t) with $B_t = Z_t$ for action zero and $B_t = 1 - Z_t$ for action one. The unused pair is independent of the past under any adaptive policy. The construction is causal, exact, and has the claimed entropy. □

The example is not offered as a physiological realization of stillness. It establishes that even exact predictive stability, frequent quiet reports, and strong stimulus discrimination do not imply a bounded or vanishing randomness requirement.

6 Empirical meaning and scope

The mathematical bridge is

experimental laws $\longrightarrow$ predictive equivalence $\longrightarrow$ finite controlled tree $\longrightarrow$ exact seed cost.

The empirical hypothesis is that reported stillness is associated with stable predictions under neutral continuation while responsiveness and wakefulness remain preserved. This must be compared against ordinary rest, mind wandering, and impaired access on held-out participants and interventions. It is not a conclusion of the quotient construction.

Recent studies of relaxed alertness [2], pure awareness [3], and mind blanking [4] motivate separating these properties. Revised cessation analyses [5] further caution against identifying a subjective state with one global complexity measure. Such measures are not automatically causal seed entropy.

The present formulation removes arbitrary latent depth scales from the mathematical bridge, but does not eliminate experimental choices. It supplies neither an established essence of stillness nor a theorem about consciousness outside spacetime. The propositions are self-contained; the imported Shannon bound and empirical interpretation have explicitly separate evidential status.

References

[1] J. Mikulík, *Finite Shannon Theorem*, companion repository. `https://github.com/jendamikulik/finite-Shannon-theorem`. A reproducible application should pin and audit the exact source commit.

[2] A. T. Duda, A. R. Clarke, and R. J. Barry (2026). Neural Oscillatory and Sympathetic Arousal Dynamics During Mindfulness Meditation: Characterizing Relaxed Alertness. *Psychophysiology* 63, e70425. `https://doi.org/10.1111/psyp.70425`.

[3] A. Chandia-Jorquera et al. (2026). Distilling the Neurophenomenological Signatures of Pure Awareness during Transcendental Meditation. *Journal of Cognitive Neuroscience* 38, 2028–2045. `https://doi.org/10.1162/JOCN.a.2626`.

[4] E. Munoz-Musat et al. (2025). Behavioral, experiential, and physiological signatures of mind blanking. *PNAS* 122, e2510262122. `https://doi.org/10.1073/pnas.2510262122`.

[5] K. Shinozuka et al. (2026). Neuroelectrophysiological correlates of extended cessation of consciousness in advanced meditators: A multimodal EEG and MEG study. *bioRxiv*, version 3, 17 October 2026. Not peer reviewed. `https://www.biorxiv.org/content/10.1101/2025.09.19.677455v3`.